

Enzymes



ENZYMES

- ❖ Enzymes are ***biochemical catalysts*** made up of globular proteins. An enzyme is always associated with a non-protein component known as **co-factor** which is tightly bonded to the enzyme.
- ❖ ***Enzymes are organic compounds*** ***protein in nature*** that speed up the rate of biochemical reactions in the body of an organism and remains ***unchanged*** at the end of the reaction.

Importance of enzymes

- ❖ The **rate** at which some **reactions** occur in the body without enzymes is too slow to sustain life. Enzymes therefore ***speed up the rate of the reaction without changing the product formed and the nature of reaction*** i.e. an enzyme cannot make a reaction that would not occur to take place and it cannot make an endothermic reaction exothermic but only ensures that products are formed in the shortest time possible.
- ❖ ***They also control metabolic processes hence promoting normal body functions***

Enzyme action

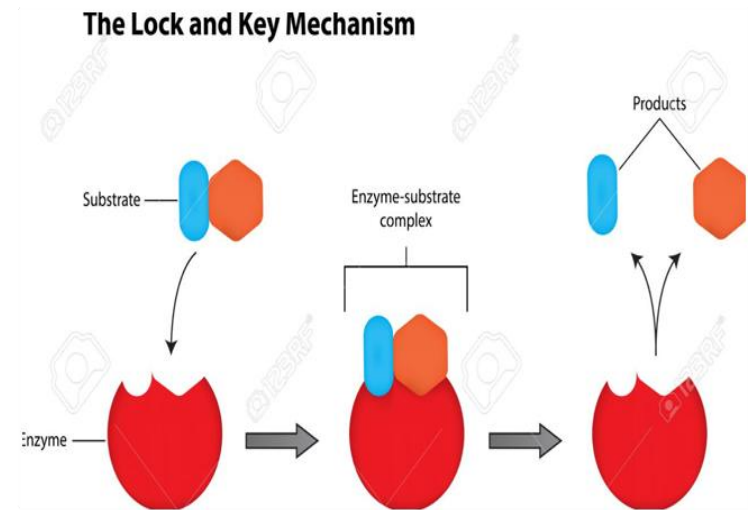
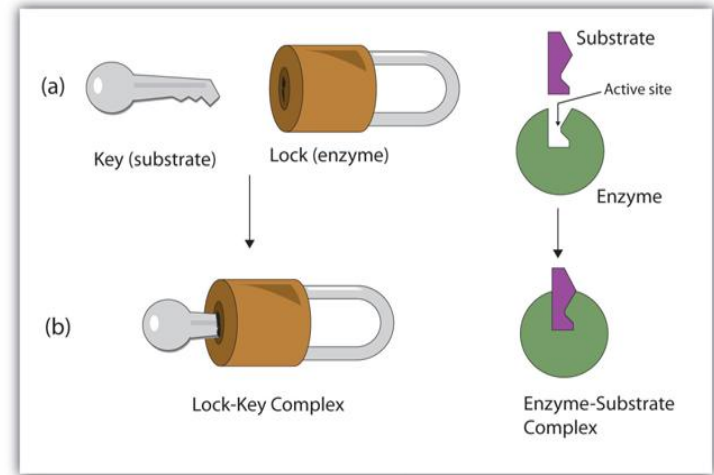
- ❖ Each enzyme has a unique surface structure which provides a precise position known as **active site**, at which the substrate can join the enzyme molecules to form an ***enzyme substrate*** complex.
- ❖ This infinite contact is maintained until the reaction is complete.
- ❖ The precise and specific fit between enzyme and substrate is sometimes compared with the **lock and key mechanisms**.

The lock and key mechanism

The widely accepted mechanism by which enzymes are known to work is the “**key and lock**” hypothesis.

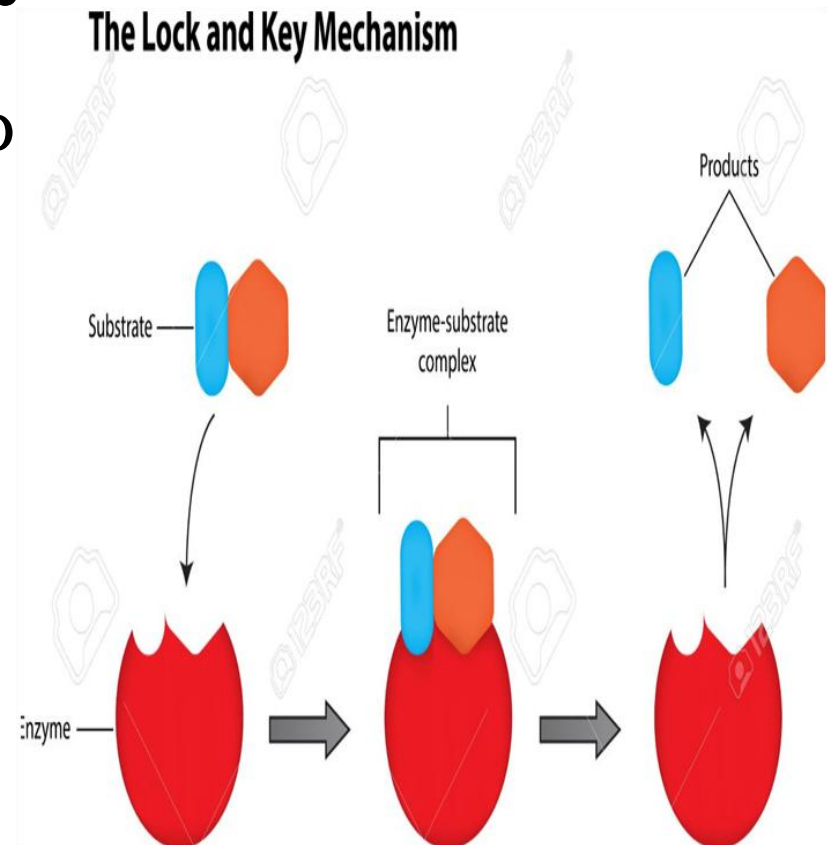
The lock and key mechanism

- The hypothesis suggests that the enzyme has a specific region known as the **active site** where the substrate fits like a key fits in a lock. The substrate must have **a complementary shape** to the active site of the enzyme.



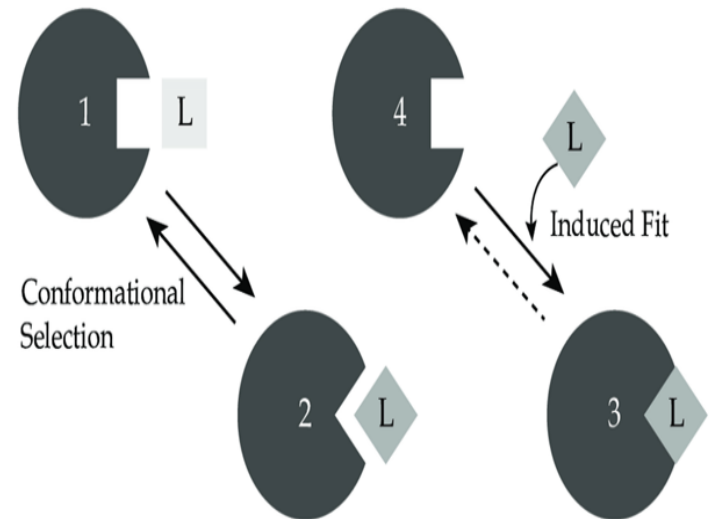
The lock and key mechanism

- ❖ In this hypothesis the **key** is **analogous** to the substrate and the **lock** to the enzyme. When the substrate combines with the enzyme, an **enzyme-substrate complex** is formed.
- ❖ This breaks down to release the products and the enzyme, which can pick other substrates.



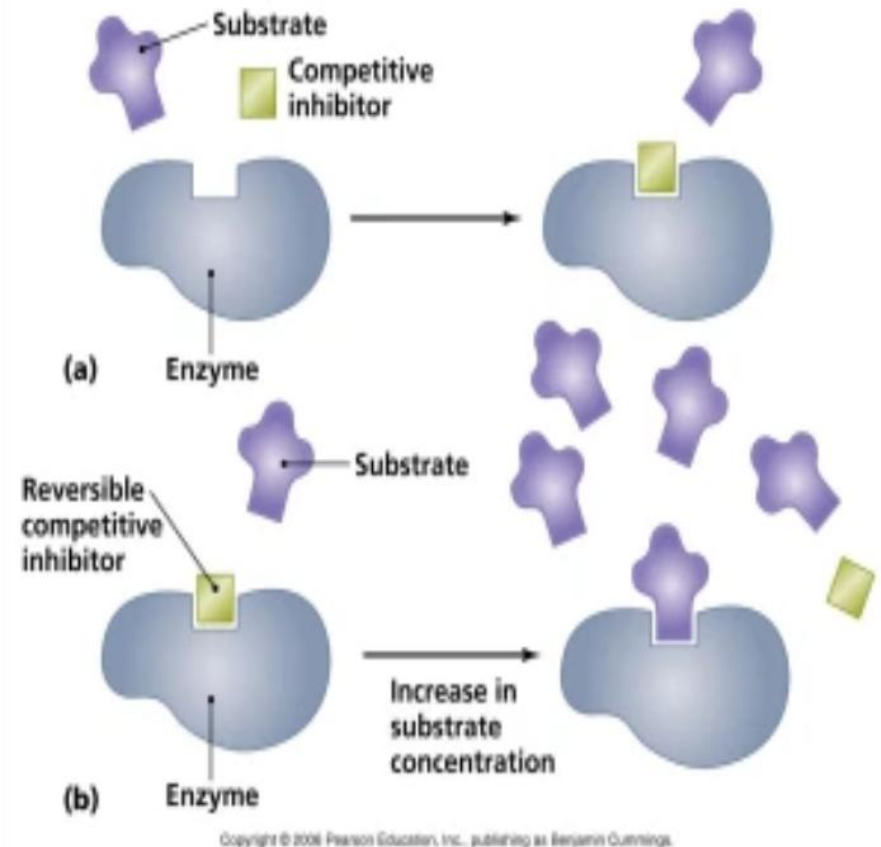
Induced fit mechanism

- ❖ It is suggested that **the active sites** may not be exactly the **same shape**. It is believed that when the substrate combines with the enzymes, it causes a small change to occur in the same shape of the same shaped enzyme molecule thereby enabling the substrate to fit more in the active site. This is known as ***induced fit***.
- ❖ The hypothesis says that when the substrate molecule enters the active site, **it causes the enzyme to change its shape so that the two molecules fit together more tightly**.



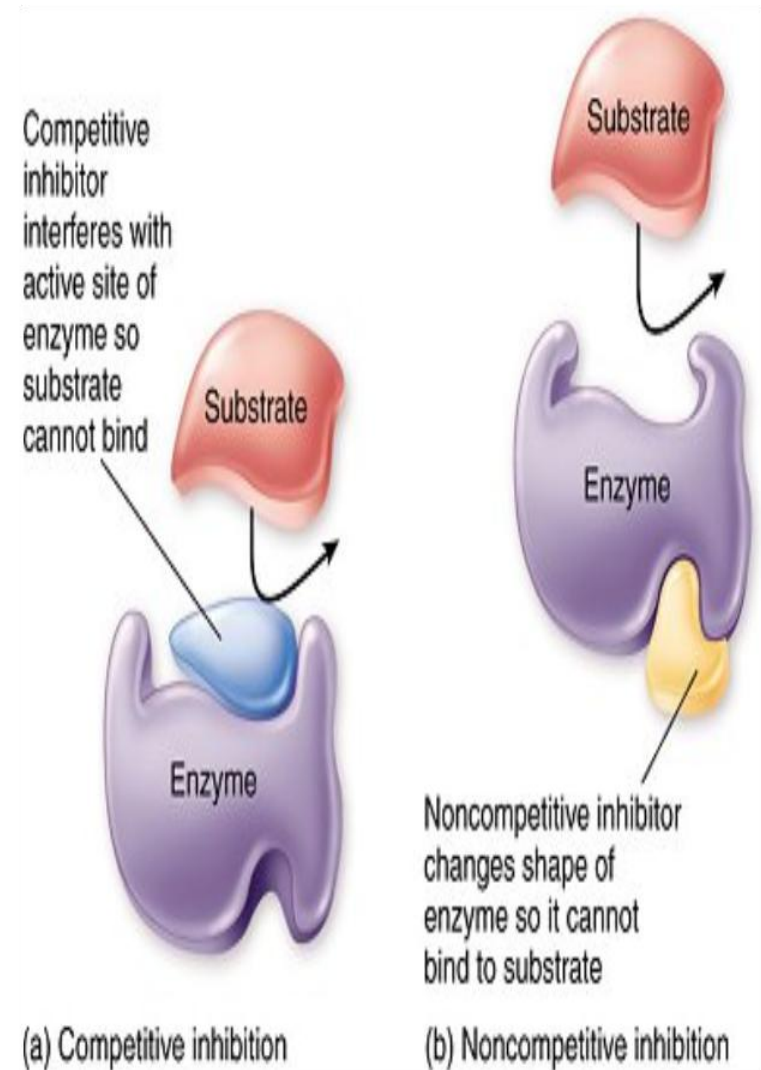
Competitive inhibition

- ✓ Enzymes may be inactivated by **substrates called inhibitors** which interfere with catalytic processes. This effect may be produced in several ways.
- ✓ Active sites at the enzyme may be blocked by the formation of the enzyme inhibitor complex. This is known as **competitive inhibition** and occurs when inhibitor molecule is structurally similar to the usual substrate of the enzyme.

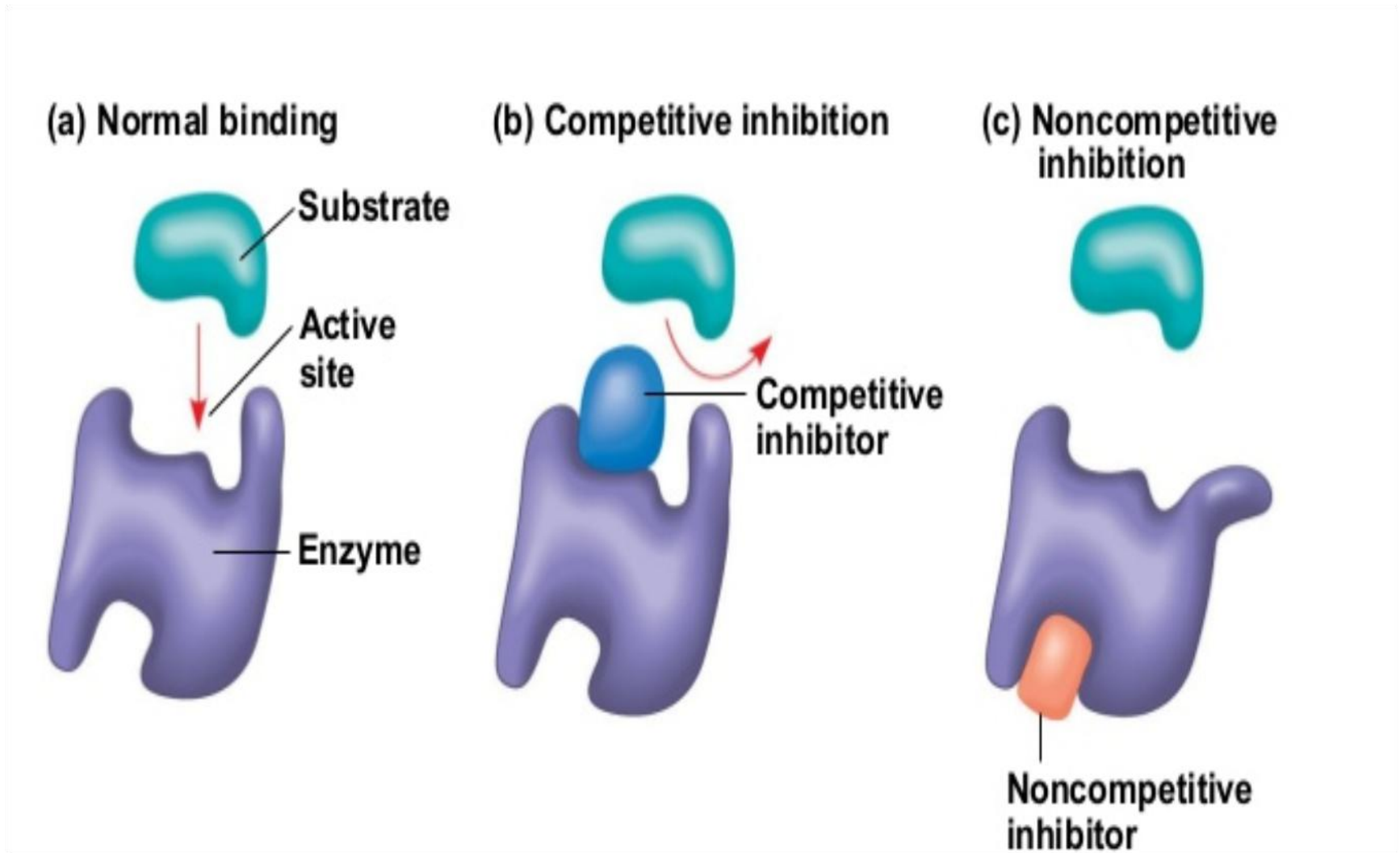


Non-competitive enzyme inhibition

- ❑ The inhibitor may **react irreversibly** with the enzyme to form an inactive non-enzymatic end product.
- ❑ The inhibitor may **alter the shape** of the enzyme at its point of activity so that the enzyme substrate complexes cannot form. This is known as **non-competitive enzyme inhibition**

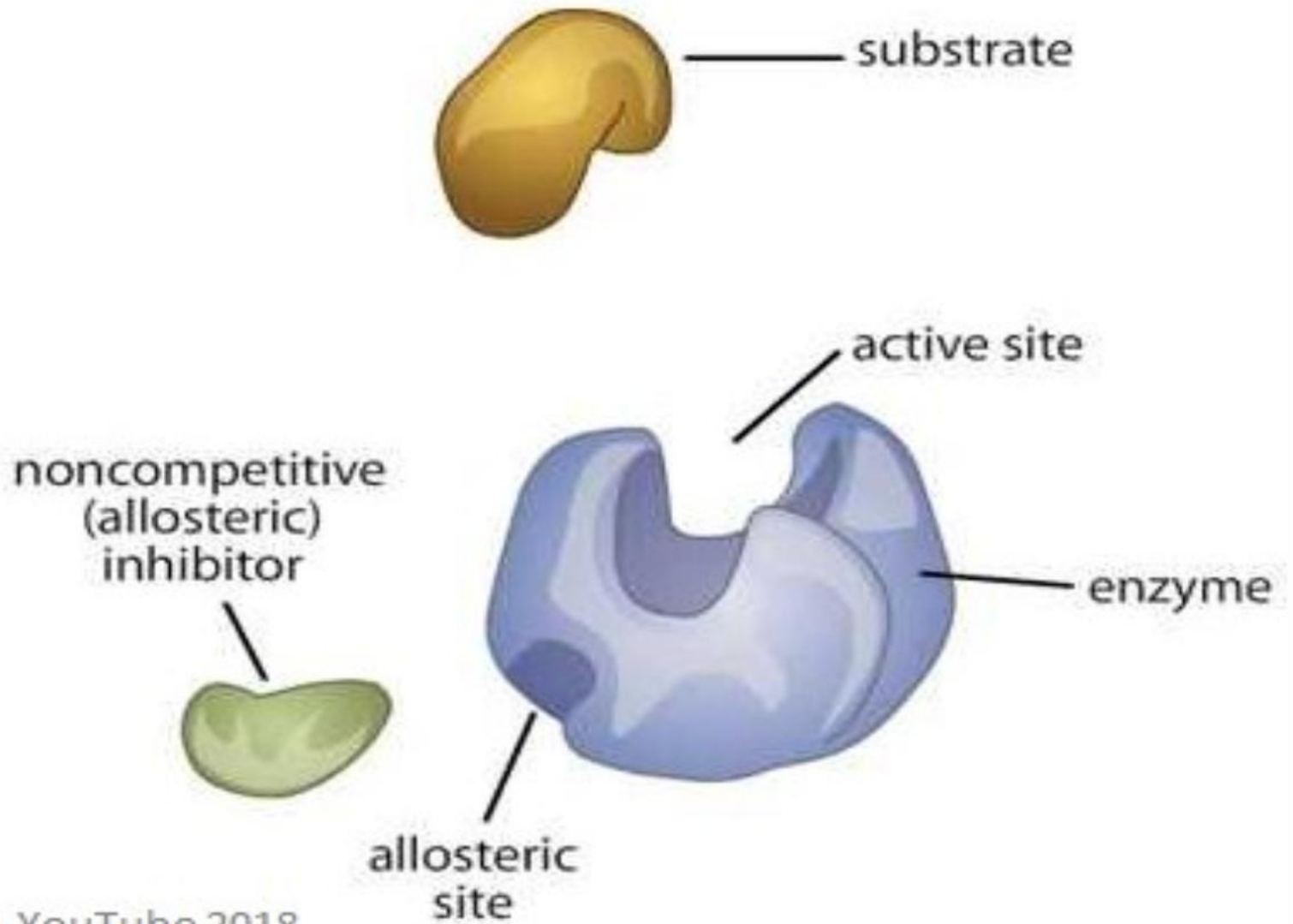


Enzyme inhibition summary



Allosteric enzymes

- ❖ These are enzymes that occur in two forms, i.e. **active** and **inactive**.
- ❖ The **inactive form** is shaped in such a way that the substrate will not fit into its active sites. Therefore for such enzymes to work, it must be transformed into the active form.
- ❖ Allosteric enzymes can be inhibited by molecules which **do not combine with the active site** but with the other parts of the enzyme. In this case, the inhibitor prevents the enzymes from changing into the active form, and substrates which bring about this are known as **allosteric inhibitors**.



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Co-factors and prosthetic groups

- ❖ In some instances an enzyme must be associated with smaller molecule in order to function properly. Such molecules are known as **co-factors**. The co-factors may be inorganic like metal ions or organic substances like nucleotides.
- ❖ **A co-enzyme** acts as a carrier for transferring chemical groups from one enzyme to another. Examples of co-enzymes include ***Nicotinamide Adenine Dinucleotide*** (NAD). Sometimes the function of the enzyme is carried out by a non-protein group, known as prosthetic group e.g. cytochrome oxidase has an iron **prosthetic**.